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Entrance Lengths for Fully Developed Laminar Flow in Eccentric Annulus

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We study the entrance length in eccentric annular geometries, where the axes of the inner and outer pipes forming the annulus are offset from each other. Such geometries are considered relevant for several industrial applications, such as drilling of wells for hydrocarbon production or geothermal energy recovery, as well as biomedical research involving annular vessels. Previous studies have shown that eccentricity increases the entrance length in annular geometries, and this has been attributed to azimuthal redistribution of fluid between the wide and narrow sides of the annulus. The present computational study aims to increase the understanding of entrance lengths in eccentric annuli for laminar flow with Reynolds numbers spanning the range from the creep limit and up to intermediate laminar values, below the regime of annular gap instabilities. We find that the entrance lengths in eccentric annuli generally increase for narrower annular gaps (higher aspect ratios). Moreover, the entrance lengths in the annuli with higher aspect ratios are also more sensitive to the degree of eccentricity compared to wider annuli (lower aspect ratios). We report empirical correlations for the dimensionless entrance length using the same form as earlier studies ($L_e = [C_0^n + (C_1 Re)^n]^{1/n}$). We find that eccentricity, which breaks the axial symmetry of the annulus and causes azimuthal flow in the entrance region, impacts the value of the exponent n*, which has been assigned the value of 1.6 in previous studies of axisymmetric conduits.* [DOI: 10.1115/1.4067129]

1 Introduction

This study addresses the entrance length required to develop the laminar velocity profile in eccentric annuli, where the axis of the inner tube is offset or decentralized with respect to the axis of the outer tube. Eccentric annular geometries are considered relevant to industrial applications such as construction of wells for oil and gas production, recovery of geothermal energy, or geological CO_2 storage [1]. Flow in eccentric annular geometries is also encountered in biomedical research in studying several types of annular vessels [2], and in fluid flow past nuclear reactor rod bundles [3]. The eccentricity of such annular geometries, i.e., imperfect centralization of the inner tube, can be the result of forces acting on the inner tube, such as gravity acting on the drill string in an inclined wellbore, or by design, such as the case for rod bundles in nuclear reactors.

Compared to fully developed flow, the flow within the entrance region of a duct is associated with an excess friction pressure drop, sometimes referred to as pressure *defect* [4], due to the extra shearing required to develop the axial velocity profile [5]. Characterization of entrance length and pressure defect within realistic process geometries can be of great importance since entrance flow will affect the total pressure drop along the pipe but also convective transport processes that are linked to the shear stress and rate-of-strain within the fluid, such as particle motion [6].

Most entrance length studies assume that the fluid enters the conduit with a uniform velocity profile enforced at the inlet cross section. Inside the conduit, fluid that is immediately adjacent to the walls will be decelerated for it to comply with the commonly invoked no-slip boundary condition, while the continuity condition ensures that the central core of the fluid accelerates. The thickness of the boundary layers that emerge adjacent to the walls grow downstream of the inlet and eventually meet as the flow becomes fully developed. Although different criteria have been used to determine the distance from the inlet and to the point where the flow is fully developed, the most frequently used is related to development of the centerline velocity, and typically defining it as the length until the centerline velocity reaches 99% of the fully developed value. This criterion is based on the assumption that the flow develops more slowly along the centerline compared to other positions in the cross section of the conduit [7].

As the flow within the entrance region of a conduit is either two- or three-dimensional, development of the velocity profile and associated friction pressure gradient is normally resolved numerically, and the entrance length is correlated to the Reynolds number, Re, for the flow. The summary of entrance length correlations for pipe geometries provided by Durst et al. showed that the normalized entrance length is found to be proportional to Re for $Re \gg 1$ and to approach a constant value in the Stokes flow limit, $Re \to 0$ [8]. In their study, a nonlinear correlation was proposed that captured the observed asymptotic behavior and also predicted the entrance length for intermediate Re within a maximum error of 3% [8]. Subsequent studies have successfully developed equivalent nonlinear correlations based on generalized Reynolds numbers for the entrance length in concentric annular geometries [9], as well as for shear-thinning or yield stress fluids in pipes [10,11]. More recently, Joshi and Vinoth examined the impacts of the uniform inlet velocity profile and the 99% centerline velocity criterion on the entrance length in channel

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and pipe geometries [7]. Their numerical results demonstrated that the uniform inlet velocity profile resulted in longer entrance lengths compared to an alternative streamtube inlet, particularly in the Stokes limit (Re $\rightarrow$ 0), and also that the velocity profile in channels develops more slowly at a transverse position, away from the centerline [7].

In addition to pipe and channel geometries, the entrance length in *concentric* (axisymmetric) annuli has also been studied previously. In a computational study that considered a significant span of annulus aspect ratios (ratios of inner to outer tube diameters), and Reynolds numbers covering several orders of magnitude, Poole found that the normalized entrance length is a function of both Reynolds number and the aspect ratio [9]. In the limit of narrow annulus, i.e., for larger diameter ratios (>0.5), the entrance length was found to approach the corresponding result for channel flow; that is, the flow in narrow concentric annulus approaches that of plane channel flow. Smaller diameter ratios were found to impact the entrance length and flow development, suggesting an increased importance of curvature on the flow [9]. As we comment below, the location of the maximum axial velocity in a concentric annulus is shifted away from the center of the annular gap and toward the inner tube—this deviation from the plane channel result increases as the effect of curvature increases.

The aim of the current study is to extend the above work to also include eccentric annular geometries, i.e., annular geometries where now the inner pipe is not perfectly centered within the outer pipe. Eccentricity of the inner pipe breaks the axial symmetry of the annular geometry and leads to a radial gap width between the pipes that vary with the azimuthal position around the annulus. This in turn impacts both the fully developed axial velocity profile and the corresponding friction pressure gradient. A derivation of the fully developed, laminar axial velocity profile in an eccentric annulus is given by Heyda [12] and Snyder and Goldstein [13], and we illustrate the impact of eccentricity on this profile in Fig. 1, based on the result in Ref. [13]. Here, the magnitude of the axial velocity profile, $\hat{u}_z$, has been normalized by the imposed bulk flow, $\hat{U} = \hat{Q}/\hat{A}$, with $\hat{Q}$ the total volumetric flux through the annular cross section of area $\hat{A}$. In Fig. 1, the ratio of inner to outer pipe diameters is set to $\gamma = \hat{D}_i/\hat{D}_0 = 0.5$, and the eccentricity is set to 0.1, 0.5, and 0.9 in the top, center, and bottom figures, respectively. As will be further explained below, the eccentricity is a dimensionless number between 0 and 1 and defined as $\varepsilon = 2\hat{c}/(\hat{D}_o - \hat{D}_i)$, where $\hat{c}$ denotes the distance between the centers of the inner and outer pipes [3]. As can be observed by comparing the three velocity profiles, fluid will flow faster in the wider side of the eccentric annulus, and this effect increases as eccentricity increases. The concentration of fluid into the wider side of the annulus is due to the greater gap width and less resistance to flow compared to the narrow side of the annulus. Thus, developing the velocity profile from a uniform profile at the inlet will involve both radial redistribution of fluid, from the walls and toward the center of the annulus, but also in the azimuthal direction from the narrow side and toward the wide side. Consequently, the flow development is fully three-dimensional within the inlet section of eccentric annuli.

While accurate entrance length correlations have been developed for both Newtonian and non-Newtonian fluids in axisymmetric ducts, such as pipes, channels, and concentric annuli, fewer studies have addressed entrance flow in the nonaxisymmetric eccentric annulus geometry. Feldman et al. developed an approximate numerical model for straight eccentric annular ducts, where a simplified relation between the transverse velocity components of the flow was assumed, and where azimuthal pressure gradients were neglected [14]. The study demonstrated that the entrance length in eccentric annuli is considerably greater than for concentric annuli with the same aspect (diameter) ratio, and that the total pressure defect in eccentric annuli is nontrivial function of both aspect ratio and eccentricity [14]. A more recent study by Velusamy and Garg relaxed the approximations of previous studies by retaining all terms in the momentum conservation equations except momentum diffusion in the axial direction [4]. A second simplification in their numerical model was to split the pressure field between an axial component and a lateral component. The study considered four different eccentricities and five annulus aspect ratios, with quantitative results that demonstrated a significant effect of eccentricity on both entrance length and pressure defect [4]. The longer entrance length for fluids entering an eccentric annulus with a uniform velocity profile is attributed to transverse (azimuthal) flow from the narrow to the wide sector of the annulus [4]. Consequently, the entrance flow is characterized by both shear in the radial direction (due to no-slip at the walls), but also transverse redistribution of fluid (due to different gap widths around the annulus).

We now relax all previous simplifications and approximations and solve the full three-dimensional conservation equations using the OPENFOAM computational platform. Furthermore, we consider entrance flow in eccentric annuli across a large range of Reynolds numbers, including both the creep (Stokes) regime and the inertial, but still laminar regime, and seek to develop empirical correlations for the entrance length. As recent experimental and computational studies [3,15] have shown that shear-driven gap instabilities occur within eccentric annular geometries at intermediate Reynolds numbers (typically Re $\leq$ 500), we limit the current study to Re < 500.

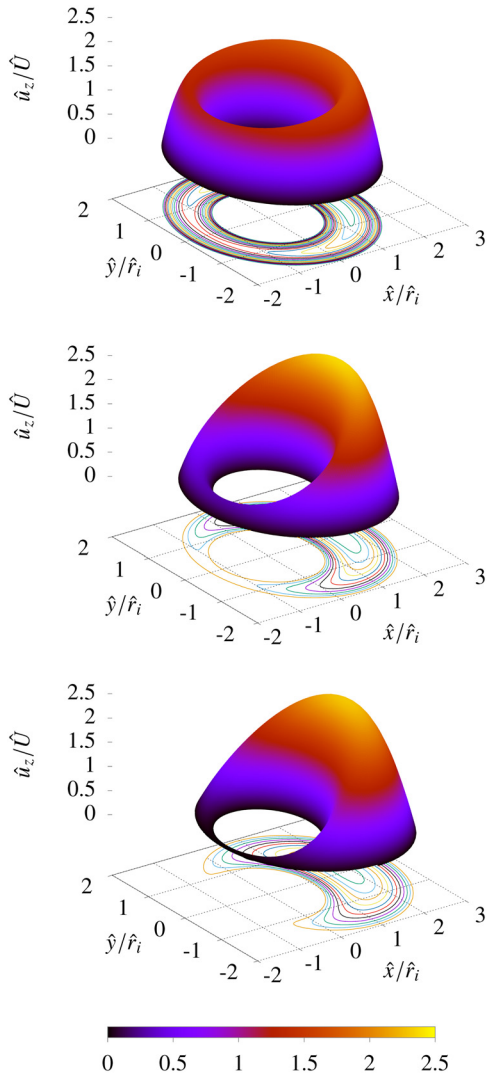


Fig. 1 Fully developed axial velocity profile in eccentric annuli (top: ε=0.1, center: ε=0.5, and bottom: ε=0.9) of aspect ratio $\gamma = D_i/D_o$=0.5. The velocity magnitude is normalized by the imposed bulk velocity, $\hat{U}$, and the cross-sectional coordinates are normalized by the radius of the inner cylinder, $\hat{r}_i$.

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2 Model Description

We assume incompressible and constant viscosity fluids, for which the governing equations are the continuity equation

$$\hat{\nabla} \cdot \hat{\mathbf{u}} = 0 \tag{1a}$$

and the momentum conservation equation

$$\hat{\rho}\left(\frac{\partial \hat{\mathbf{u}}}{\partial \hat{t}} + (\hat{\mathbf{u}} \cdot \hat{\nabla})\hat{\mathbf{u}}\right) = -\hat{\nabla}\hat{p} + \hat{\mu}\hat{\nabla}^2\hat{\mathbf{u}} \tag{1b}$$

Here, $\hat{\mathbf{u}}$ is the velocity vector of the fluid, $\hat{\rho}$ and $\hat{\mu}$ denote the (assumed constant) mass density and shear viscosity of the fluid, respectively, and $\hat{p}$ is the pressure. Here, and in the following, dimensional quantities are identified by a caret, $\hat{}$. The assumption of incompressibility is considered appropriate for entrance flow of liquids, as the total pressure variation over the domain is relatively small. It is noted that the compressibility effect can be significant in certain applications with very long conduits. However, the assumption is considered valid in this application, since the model in question is reasonably short. The governing equations (1*a*) and (1*b*) are supplemented by boundary conditions for velocity and pressure. We assume no-slip boundary conditions at the inner and outer walls of the annulus and assume a uniform axial velocity profile at the inlet. The flow is assumed to be fully developed before reaching the end of the geometry, so we fix the axial gradient of the velocity profile to be zero at the outlet cross section. The pressure is assumed fixed at the outlet, and the inlet pressure is determined using a Neumann boundary condition of zero axial gradient.

Before proceeding, we comment on the assumptions of constant viscosity and no-slip boundary condition at the solid walls. Industrial fluids are often multicomponent fluids, such as emulsions, and/or may consist of dispersed colloidal or noncolloidal particles. These fluids are generally non-Newtonian and may exhibit a combination of shear-thinning, yield stress behavior and viscoelasticity and/or thixotropy [16]. As the viscosity of non-Newtonian fluids is often assumed to be a function of the local shear-rate, the viscosity of such fluids will vary both in the axial direction and over the eccentric annulus cross section. Further, the final velocity profile itself may exhibit a nontrivial combination of static, flowing, and even partially sheared "plug" regions in eccentric annuli [17]. In addition, the hydrodynamic forces associated with the flow along the inlet section will affect the microscopic structure of dispersions and multicomponent industrial fluids [18], potentially resulting in viscoelastic and/or significant thixotropic responses that can affect the dynamics of flow development [19]. The current study is based on a simple (constant viscosity) fluid model, and it is noted that realistic industrial fluids may yield significantly different results. Finally, we consider the assumption of a no-slip boundary condition at the walls of the eccentric annulus to be appropriate for simple fluids in typical fluids engineering settings. Industrial (multicomponent) fluids may exhibit migration of the dispersed phase and/or suspended particles away from the walls, which would result in depleted wall layers and apparent slip phenomena [20].

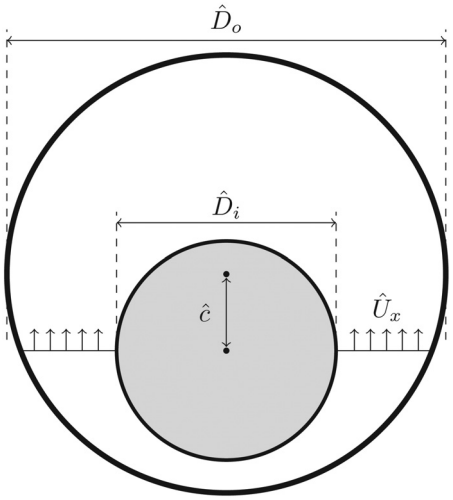


Fig. 2 The eccentric annulus cross section is defined by the inner and outer tube diameters, $\hat{D}_i$ and $\hat{D}_o$, respectively, and the transverse offset distance $\hat{c}$ between the center of the tubes. The cross section illustrated here has an aspect ratio $\gamma = \hat{D}_i/\hat{D}_o = 0.5$ and eccentricity $\varepsilon = 2\hat{c}/(\hat{D}_o - \hat{D}_i) = 0.7$. Also illustrated is the horizontal plane through the annulus used to calculate the magnitude of the transverse flow, $\hat{U}_x$. This plane coincides with the center of the inner tube.

We consider eccentric annular geometries, such as that illustrated in Fig. 2, defined by inner and outer diameters $\hat{D}_i$ and $\hat{D}_o$, respectively, and define the hydraulic diameter of the annulus as $\hat{D}_H = \hat{D}_o - \hat{D}_i$, equivalent to four times the annulus cross-sectional area divided by the wetted perimeter of the annulus [5]. As illustrated in Fig. 2, the offset between the axis of the inner and outer pipes for eccentric annuli is denoted $\hat{c}$. We assume this offset to be constant along the length of the annular geometry and assume smooth inner and outer pipes, with perfectly circular cross sections.

Also shown in Fig. 2 is the plane through which we evaluate the transverse bulk velocity, $\hat{U}_x$. This quantity can be useful for comparing the magnitude of the imposed (axial) bulk flow and the transverse bulk velocity that is due to the eccentric orientation of the inner tube. This plane, which is horizontal in Fig. 2, is defined so that it coincides with the center of the inner tube in all of the annular geometries considered in this study.

Based on the model description provided above, we identify the following dimensionless numbers. The annular geometry is characterized by its aspect ratio and the eccentricity of the inner tube. The aspect ratio, taken as $\gamma = \hat{D}_i/\hat{D}_o$, is a measure of the "narrowness" of the annular gap, while the eccentricity is the ratio of tube axis offset to radial distance between inner and outer walls in the concentric case, i.e., $\varepsilon = 2\hat{c}/(\hat{D}_o - \hat{D}_i)$. Defined as such, both γ and ε take values in the interval from 0 to 1. From the momentum equation (1*b*), we identify the Reynolds number $\mathrm{Re} = \hat{\rho}\hat{U}\hat{D}_H/\hat{\mu}$ as dimensionless number. The Reynolds number is based on the hydraulic diameter of the annulus, $\hat{D}_H$, and the imposed axial (bulk) velocity, $\hat{U}$. From dimensional analysis [5], we therefore expect the dimensionless entrance length, $L_e = \hat{L}_e/\hat{D}_H$, to be completely determined by an unknown function f of the Reynolds number and the geometry of the annulus, viz.,

$$L_e = \hat{L}_e/\hat{D}_H = f(\mathrm{Re}, \gamma, \varepsilon) \tag{2}$$

with $\hat{L}_e$ being the (dimensional) axial length from the inlet of the annulus and to the downstream position where the velocity profile is identical to the fully developed profile. In this study, the entrance length is defined as the axial position downstream of the inlet of which the maximal velocity in the cross section has reached 99% of the maximal velocity of the fully developed flow [21].

The form of $f(\mathrm{Re}, \gamma, \epsilon)$ can vary depending on the studied flow and geometric parameters. An empirical correlation that has proven successful for axisymmetric conduits, such as pipes, channels, and concentric annuli, is

$$L_e = \left[C_0^n + (C_1\mathrm{Re})^n\right]^{1/n} \tag{3}$$

In Eq. (3), the parameter C_0 represents the entrance length in the creep flow, or Stokes flow, limit where $\mathrm{Re} \to 0$, while the term $C_1\mathrm{Re}$ represents the entrance length in a laminar and inertial range, where $C_1\mathrm{Re} \gg C_0$. The exponent n aims to better capture the entrance lengths in the intermediate range between these two limits, with $n = 1.6$ being a typical value for the above-mentioned geometries [8,9]. The nonlinear relation in Eq. (3) was proposed by Churchill and Usagi, who found this form to be a robust correlation for a range of different transport processes [22]. In Eq. (3), the parameter C_0 represents the entrance length in the creep flow, or Stokes flow, limit

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where $\mathrm{Re} \to 0$, while the term $C_1\mathrm{Re}$ represents the entrance length in a laminar and inertial range, where $C_1\mathrm{Re} \gg C_0$.

3 Numerical Setup

Our numerical experiments were conducted using the computational fluid dynamics software OPENFOAM with the semi-implicit method for pressure linked equations-consistent algorithm for pressure–velocity coupling. Second-order discretization schemes were used for all spatial terms, and a convergence criterion of 10^{-7} was used for the residuals of velocity and pressure. The simulations were carried out on three different diameter ratios $\boldsymbol{\gamma} = \{0.2, 0.5, 0.8\}$, each representing a low, medium, and high ratio, respectively. Within the domain of well construction, we note that these aspect ratios are considered relevant for the eccentric annular flow outside a drill string (0.2, 0.5) and along the annular space between a casing string and the rock formation (0.8) [23]. For each γ, six different values for the eccentricity were used: $\boldsymbol{\varepsilon} = \{0.0, 0.1, 0.3, 0.5, 0.9\}$ which resulted in a total of 18 different geometries.

The Reynolds number range was chosen so that the characteristic curve of the entrance length was sufficiently captured in the range between creeping flow and up to the limit of the presence of shear-driven gap instabilities [3,15]. This resulted in 16 different Reynolds numbers with the values: $\mathbf{Re} = \{0.1, 1, 5, 10, 25, 50, 75, 100, 150, 200, 250, 300\}$ and a total of 288 simulations. Three different sets of inlet cell sizes and domain lengths were used for three different sets of Reynolds numbers, grouped as low: $\mathbf{Re}_L = \{0.1, 1, 5, 10, 25, 50\}$, medium: $\mathbf{Re}_M = \{75, 100, 150\}$, and high: $\mathbf{Re}_H = \{200, 250, 300\}$. The three groups used an initial axial cell length at the inlet of $\Delta\hat{L}_{\min}/\hat{D}_o = \{0.002, 0.006, 0.01\}$ and a total domain length of $\hat{L}/\hat{D}_o = \{8, 30, 60\}$ for the low, medium, and high groups of Reynolds numbers, respectively. These groups were defined to minimize the number of cells while complying with the varying requirements of the different Reynolds numbers. The total domain length was chosen to be sufficiently large to guarantee fully developed conditions at the outlet for the different groups of Reynolds numbers. In the axial direction, a finer resolution was required at the inlet compared to further downstream in the annulus to capture the gradients in the development region sufficiently. All the groups used a growth rate from the inlet of $g = 1.1$, of which the cell size grew in the axial direction until reaching the maximal axial cell length $\Delta\hat{L}_{\max}/\hat{D}_o$. An illustration of the numerical grid used in the study is shown in Fig. 3 with an eccentricity of $\varepsilon = 0.5$ and an aspect ratio of $\gamma = 0.5$. The illustration shows a grid with $N_r = 35$ and $N_\theta = 200$ number of divisions in the radial and azimuthal directions, respectively. Additionally, the illustration shows a grid with a maximal axial cell length of $\Delta\hat{L}_{\max}/\hat{D}_o = 0.4$.

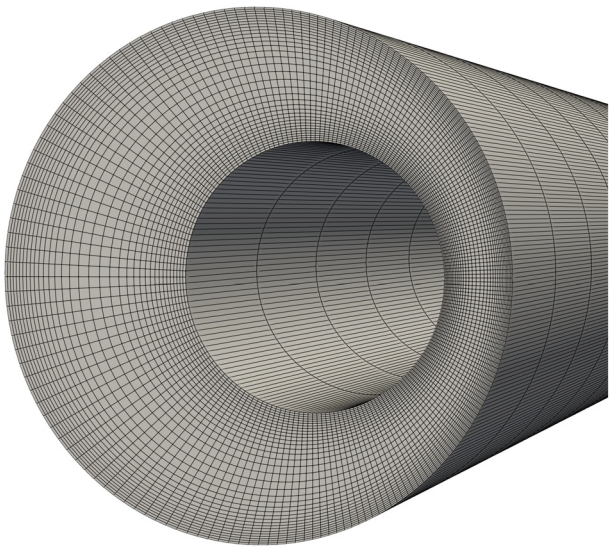


Fig. 3 Illustration of the grid for γ=0.5 and ε=0.5 with N_r = 35, N_θ=200, and $\Delta\hat{L}_{\max}/\hat{D}_o$=0.4

The entrance length was automatically determined for each simulation by identifying the axial position where the maximum velocity in the cross section reaches 99% of the corresponding fully developed value. The fully developed velocity was calculated using the result of Snyder and Goldstein [13] which, for completeness, is reproduced in the Appendix. This criterion is chosen to be consistent with previous studies of development flow in eccentric annuli [4,14], as well as more recent investigations of pipe, channel, and concentric annular geometries. We do acknowledge, however, that other locations within the eccentric annular cross section may develop more slowly than the location of the maximum velocity; alternative, integral-based criteria may be assessed for future studies of development flow in this nonaxisymmetric geometry [7].

With the entrance lengths determined based on the maximum velocity criterion specified above, the correlation constants C_0 and C_1 in Eq. (3) were identified using the entrance length found for the lowest ($\mathrm{Re} = 0.1$) and the highest ($\mathrm{Re} = 300$) Reynolds numbers, respectively. The exponent n in Eq. (3) was finally calculated for each annular geometry by minimizing the sum of squared residuals

$$\mathrm{RMS}(n) = \sum_{\mathrm{Re}\in\mathbf{Re}} \left(\frac{L_e(n, \mathrm{Re}) - L_{\mathrm{CFD}}(\mathrm{Re})}{L_{\mathrm{CFD}}(\mathrm{Re})} \right)^2 \tag{4}$$

where L_{CFD} is the dimensionless entrance length extracted from each set of simulation.

3.1 Grid Refinement Study. Because of the large span of simulations in this study, the numerical resolution in the radial and azimuthal directions was fixed across all simulations. A refinement study was conducted for the highest Reynolds number ($\mathrm{Re} = 300$) with the geometry corresponding to the thinnest and most eccentric ($\varepsilon = 0.9$, $\gamma = 0.8$). This combination of governing parameters was picked based on having the highest requirement for the grid resolution in the cross section. A summary of the grid refinement study of the cross section is illustrated in Table 1. The maximal axial resolution was initially fixed to $\Delta\hat{L}_{\max}/\hat{D}_o = 0.2$, which is equivalent to the value used in the study by Lamarche-Gagnon and Tavoularis [15]. The total length of the domain was fixed to $\hat{L}/\hat{D}_o = 60$. A relative error of the maximal fully developed velocity is defined as

$$E_r = \frac{\hat{U}_c - \hat{U}_a}{\hat{U}_a} \tag{5}$$

in which $\hat{U}_c$ is the calculated maximal velocity at the outlet plane, and $\hat{U}_a$ is the maximal fully developed velocity from the semi-analytical solution presented in the Appendix. For the previously described parameters of the grid refinement study, the fully developed semi-analytical solution resulted in $\hat{U}_a = 0.7267\,\mathrm{m\,s^{-1}}$.

Based on the presented cross section grid refinement study, the "M2" mesh size was chosen for the remainder of the study, with $N_r = 35$ and $N_\theta = 200$. The resolution was chosen as it provided a

Table 1 Grid refinement study of the cross section with varying values of the radial resolution (N_r) and the azimuthal resolution (N_θ)

Mesh size	N_r	N_θ	$\hat{U}_c$ (m s^{-1})	E_r
M1: 3500	35	100	0.7184	−0.0114
	25	120	0.7227	−0.0055
M2: 7000	50	120	0.7227	−0.0055
	70	100	0.7188	−0.0109
	35	200	0.7244	−0.0032
M3: 14,000	70	200	0.7248	−0.0026
	35	400	0.7259	−0.0011

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Table 2 Grid refinement study of the axial cell length with fixed cross section resolution

Axial cell length	$\hat{U}_c$ (m s^{-1})	E_r
$\Delta\hat{L}_{ax}/\hat{D}_o = 0.2$	0.7244	−0.0032
$\Delta\hat{L}_{ax}/\hat{D}_o = 0.4$	0.7244	−0.0032
$\Delta\hat{L}_{ax}/\hat{D}_o = 0.6$	0.7244	−0.0032

sufficient balance between accuracy and computational speed. A grid refinement study was also conducted on the axial cell length to investigate whether a larger axial cell size could provide sufficient accuracy than the one used for the initial grid refinement study. The results of the study are shown in Table 2.

The results indicate a very low sensitivity of the cell length in the axial direction in the fully developed region. For the remainder of the study, the maximal axial cell length used across all studies is fixed at $\Delta\hat{L}_{max}/\hat{D}_o = 0.4$.

The axial cell length in the development region exhibited higher sensitivity due to the steep axial gradient close to the inlet. The initial cell lengths at the inlet $\Delta\hat{L}_{min}/\hat{D}_o$ that were presented in Sec. 3 were obtained based on the lowest Reynolds number in each group of Reynolds numbers. The cell lengths were chosen empirically, where the sensitivity of the extracted entrance length of each test was used as a benchmark. The initial cell lengths were picked at the largest value where no significant change in entrance lengths could be detected. Additionally, the entrance length correlation function in Eq. (3) was also used as a benchmark for the grid refinement. In the case that the function did not accurately predict the simulated entrance lengths across the whole range of Reynolds numbers, the initial axial cell lengths were refined where deemed necessary. Numerical Setup

3.2 Validation Case: Concentric Annulus. The numerical model was first validated for the concentric annulus ($\varepsilon = 0$) and using results for $\gamma = 0.5$ and $\gamma = 0.8$, published by Poole [9]. The analytical result for the fully developed velocity profile was used to determine the entrance length in this case (see, e.g., Ref. [5]), which occurs at a radial position that is not at the center of the annular gap but closer to the inner tube

$$\hat{R}_{max,c} = \frac{\hat{D}_i + \hat{D}_o}{4}\sqrt{\frac{2(1-\gamma)}{(1+\gamma)|\ln\gamma|}} \leq \frac{\hat{D}_i + \hat{D}_o}{4} \tag{6}$$

The square root in this expression is less than 1 for any $\gamma < 1$, and it approaches 1 as $\gamma \to 1$. Consequently, for an infinitely narrow annular clearance, the peak magnitude resides at the center of the annulus; otherwise, it will be located closer to the inner tube wall. Figure 4 compares the normalized entrance length results from the current study to results from Ref. [9] and illustrates good agreement with the extracted results. Shown in these plots as a solid line is the correlation function from Eq. (3). Here, $n = 1.6$, as found by Ref. [9], and with C_0 and C_1 parameters given as per Table 3. We find satisfactory quantitative agreement between the 2D numerical model of that previous study [9] and the current fully 3D simulation model.

4 Results

4.1 Pressure Distribution and Azimuthal Bulk Flow. As pointed out by, e.g., Velusamy and Garg, eccentricity is found to significantly increase the entrance length compared to a concentric annulus of the same aspect ratio, and this has been attributed to the redistribution of fluid between the narrow and the wide side of the eccentric annulus [4].

To illustrate the transverse flow mechanism within the development region, the normalized pressure distribution across the eccentric annular cross section are shown for $\mathrm{Re} = 200$ in Figs. 5 and 6 for different downstream positions. These cross sections are characterized by significant eccentricity ($\varepsilon = 0.7$) and with either low or high aspect ratios ($\gamma = 0.2, 0.8$). To enhance the visualization of pressure differences between the wide and narrow sides, we here plot a normalized pressure, p_z, measured relative to the mean pressure $\langle\hat{p}(\hat{z})\rangle$ of each cross section

$$p_z = \frac{\hat{p}(\hat{z}) - \langle\hat{p}(\hat{z})\rangle}{\langle\hat{p}(\hat{z})\rangle} \tag{7}$$

Due to the azimuthal symmetry of the eccentric annulus, only one half of the cross sections are shown.

The cross-sectional plots in Figs. 5 and 6 are all acquired from the entrance regions for these two cases, where a clear pressure differential exists between the wide and narrow sides of the annulus. The pressure difference is linked to flow from the narrow side (high pressure) to the wide side (lower pressure), which is characteristic of the eccentric (nonaxisymmetric) geometry of the annulus. We note that the low- and high-pressure regions span the entire annulus gap at these locations, producing mainly azimuthal flow within the cross section. The larger difference is observed for the high-aspect ratio

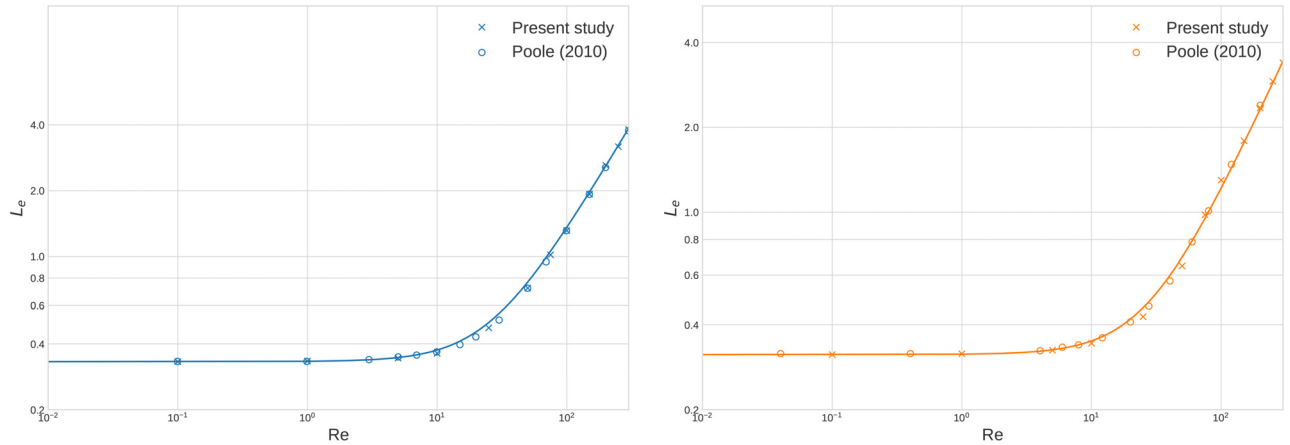


Fig. 4 Validation of computational method used in the present study by comparing dimensionless entrance lengths to those obtained by Poole [9] for concentric annuli. The solid line plots the correlation equation in Eq. (3) with $n = 1.6$. (*a*) γ=0.5 and (*b*) γ=0.8.

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Table 3 Entrance length correlation constants C_0, C_1, and n defined in Eq. (3)

	$\gamma = 0.2$			$\gamma = 0.5$			$\gamma = 0.8$		
ε	C_0	C_1	n	C_0	C_1	n	C_0	C_1	n
0.0	0.3664	0.0146	1.7852	0.3319	0.0127	1.8993	0.3137	0.0113	1.7282
	—	0.0148[a]	—	0.3282[b]	0.0123[a], 0.0123[b]	—	0.3143[b]	0.0120[a], 0.0120[b]	—
0.1	1.0989	0.0810	1.6141	2.3635	0.1099	1.7282	6.0641	0.1075	1.7282
0.3	1.2673	0.0946	1.7852	2.6117	0.1434	2.0705	7.1749	0.1662	2.1846
0.5	1.1348	0.0914	1.8993	2.3572	0.1627	2.2987	6.7029	0.2291	2.5839
	—	0.0886[a]	—	—	0.1478[a]	—	—	0.2064[a]	—
0.7	0.9407	0.0891	1.8423	2.0268	0.1767	2.3557	5.8982	0.2949	2.8691
0.9	0.8031	0.0889	1.7852	1.7762	0.1909	2.1846	5.4561	0.3738	3.0403

[a]Result extracted from Velusamy and Garg [4].
[b]Result extracted from Poole [9].

annulus, $\gamma = 0.8$, which will direct fluid toward the wide side of the annulus.

To further examine the transverse redistribution of fluid within the entrance region, we plot the normalized azimuthal bulk flow, $\hat{U}_x/\hat{U}$, in Fig. 7(*a*). We recall that $\hat{U}_x$ is taken as the mean, or bulk transverse flow through a plane that intersects the center of the inner tube, and that is perpendicular to the symmetry plane of the eccentric annulus, as shown in Fig. 2. In Fig. 7(*a*), we plot the axial variation of the bulk transverse flow for the same conditions as in Figs. 5 and 6 and also an intermediate aspect ratio, $\gamma = 0.5$ (*a*). Figure 7(*b*) shows a contour plot of the velocity component $\hat{u}_x/\hat{U}$ extracted from the same plane as $\hat{U}_x/\hat{U}$ to further illustrate the development of the azimuthal flow.

We note that the axial position is normalized by the entrance length for each of these cases. The figure demonstrates a clear trend for increased azimuthal bulk velocity in the narrow (high-aspect ratio) annulus ($\gamma = 0.8$), which aligns with the stronger pressure variation noted in Fig. 6; the azimuthal bulk velocity is found to be greater than 25% of the imposed bulk velocity in this case. We note a reduction in the peak azimuthal bulk velocity as the aspect ratio decreases to $\gamma = 0.5$ and 0.2, and also that the magnitude of the azimuthal bulk velocity decreases rapidly downstream of the inlet at this value of Re and eccentricity.

In Fig. 8, we consider the more modest eccentricity of $\varepsilon = 0.3$ and immediately observe a reduction in the magnitude of the azimuthal bulk velocity compared to the case of $\varepsilon = 0.7$, and that significant azimuthal flow persists further downstream, especially for the higher-aspect ratio annulus ($\gamma = 0.8$) shown in Fig. 8.

In both Figs. 8 and 7, we note that the peak value of the azimuthal bulk flow occurs further downstream for the high-aspect ratio annulus. The larger velocity and the later occurrence of the maximum value may be attributed to both a narrower radial gap (increased wall drag) and a larger, azimuthal distance between the wide and narrow sides.

4.2 Entrance Lengths. We next address the entrance length in the eccentric annulus and plot the normalized entrance length, L_e, as function of Re in Fig. 9 for $\gamma = 0.5$ and 0.8 and for eccentricities of 0.5 and 0.9.

In addition to showing the entrance lengths from the simulations as filled points, the plot also includes variations of the correlation function from Eq. (3) in the same plot. The dashed curve shows the linear form of the correlation function with $n = 1$, while the dash-dotted curve shows the nonlinear correlation with $n = 1.6$, as suggested from earlier studies for pipe and channel geometries [8]. Using $n = 1.6$ is seen to affect the transition from the creep regime, $L_e \sim C_0$, to that of higher Reynolds numbers, $L_e \sim C_1 \mathrm{Re}$, and results in a significant improvement of the entrance length prediction compared to the linear function, $n = 1$. Finally, the solid curve shows the correlation function with the exponent calculated in the current study using Eq. (4), which results in an even better accuracy for the entrance length prediction. For the cases studied here, the improved correlations with new exponents correspond to values $n > 1.6$. It can be seen from the plots in Fig. 9 that the new exponents are especially important to capture the entrance lengths in the regime between the creep limit and the upper laminar regime, which has not yet been assessed in earlier studies.

A summary of the correlation constants for all the simulated geometries included in the present study is shown in Table 3, which also includes comparable results from earlier studies of concentric annuli [9] and eccentric annuli [4]. Further, Fig. 10 shows a plot illustrating how the correlation constants C_0 and C_1 vary over the range of eccentricities and across the different aspect ratios.

As shown by the summary of correlation constants in Table 3, eccentricity increases the entrance length compared to the concentric case for all three aspect ratios considered. In the limit that $\mathrm{Re} \to 0$, the evolution of the velocity profile is governed by the balance of viscous diffusion and pressure differences, and the dimensionless entrance length is given by C_0. For a fixed eccentricity, ε, we note that the magnitude of this coefficient increases with increasing aspect ratio. For the eccentric cases, we find that the ratio $C_0(\varepsilon, \gamma = 0.5)/C_0(\varepsilon, \gamma = 0.2) \approx 2.1 - 2.2$, while

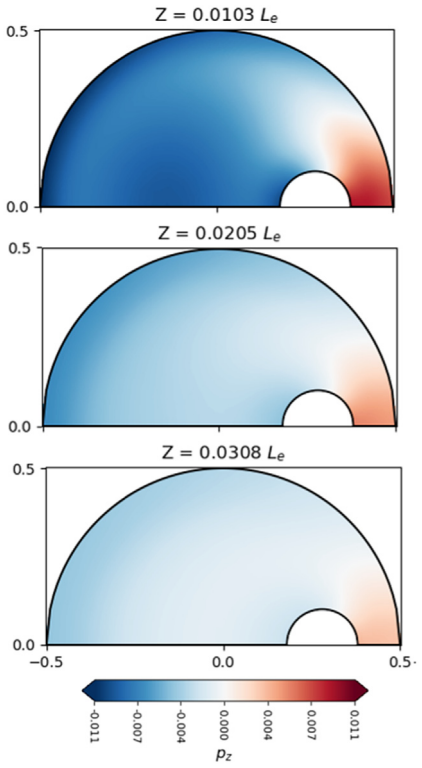


Fig. 5 Cross-sectional variations in normalized pressure, p_z, (defined in Eq. (7)) for Re = 200, γ=0.2, and ε=0.7, sampled at increasing distances downstream of the inlet

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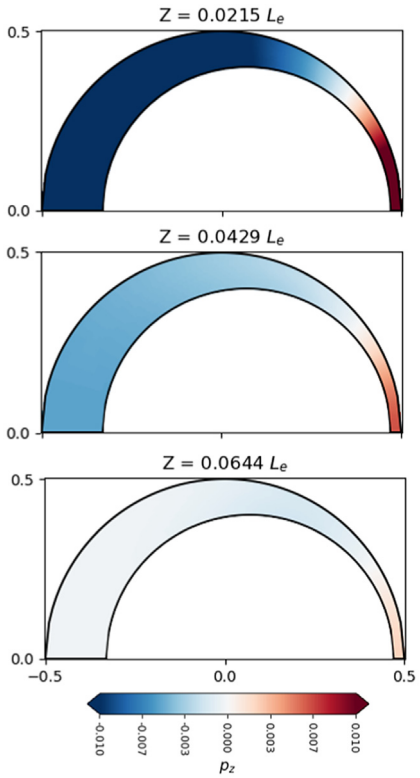


Fig. 6 Cross-sectional variations in normalized pressure, p_z, (defined in Eq. (7)) for Re = 200, γ=0.8, and ε=0.7, sampled at increasing distances downstream of the inlet

$C_0(\varepsilon, \gamma = 0.8)/C_0(\varepsilon, \gamma = 0.2) \approx 5.6 - 6.3$. We may connect this trend to the variation in mean circumference or azimuthal distance between the wide and narrow sectors as function of γ: Taking $\hat{S} = \pi(\hat{R}_i + \hat{R}_o)/2$ as this mean azimuthal distance, we have that $S = \hat{S}/\hat{D}_H = (\pi/4)(1+\gamma)/(1-\gamma)$. Thus, $S(\gamma{=}0.5)/S(\gamma{=}0.2){=}2$, while $S(\gamma{=}0.8)/S(\gamma{=}0.2){=}6$, which are similar to the corresponding ratios of C_0 and entrance length found above. In other words, the dimensionless entrance length in the creep regime may be linked to the dimensionless azimuthal distance between the wide and narrow sides of the annulus. The dependence of C_0 with ε at fixed γ suggests a nonmonotonic trend, with the longest entrance length at ε=0.3 in the creep regime. We note that C_0 appears to vary stronger with γ than it does with ε for the eccentric cases, as is also illustrated in the plot from Fig. 10(*a*). Table 3 also shows that the perfectly concentric cases have smaller variations in C_0 with γ: Indeed, for the concentric cases, C_0 decreases as γ increases which may be linked to the reduced importance of curvature as γ increases and the annulus becomes narrower, and more similar to a plane channel.

For the laminar, inertial regime at higher Reynolds numbers, i.e., the regime where entrance length is governed by C_1Re, we note from Table 3, and also the plot from Fig. 10(*b*), that C_1 again increases with increasing γ at a fixed eccentricity. Increasing γ results in a narrower annulus and a greater azimuthal distance between wide and narrow sides, as discussed above.

We also note that, for a fixed γ, that the value of C_1 is nearly linear in ε: For $\gamma = 0.2$, C_1 is nearly independent of ε, while $C_1(\varepsilon, \gamma = 0.5) \sim 0.08\varepsilon$ and $C_1(\varepsilon, \gamma = 0.8) \sim 0.3\varepsilon$. Thus, the entrance length for the narrower annulus ($\gamma = 0.8$) is more sensitive to eccentricity than the wider annuli in this inertial regime.

The accuracy of the correlation function in Eq. (3) in the intermediate region between the creep flow limit and the inertial regime is dependent on the size of the exponential coefficient n, as illustrated in Fig. 9. The values of the exponent summarized in Table 3 illustrate that the earlier studies from pipe flow and channel flow with $n = 1.6$ give a sufficient correlation for the lowest aspect ratios in general ($\gamma = 0.2$) and also generally for concentric and low eccentric values ($\varepsilon \leq 0.1$). However, for the higher aspect ratios with moderate to high eccentricities ($\varepsilon \geq 0.3$), this value for n does not accurately capture the entrance lengths for the whole range of Reynolds numbers.

5 Summary

Laminar flow in annular geometries was simulated for three different aspect ratios and six different eccentricities spanning Reynolds numbers ranging from the creep limit up to the inertial regime, in the absence of gap instabilities. Entrance lengths were extracted for each geometry over the range of Reynolds numbers, and coefficients for nonlinear entrance length correlation functions were calculated. The results illustrate a significant increase in entrance lengths when comparing results for concentric and eccentric annuli. For eccentric annuli in the creep limit, a relation

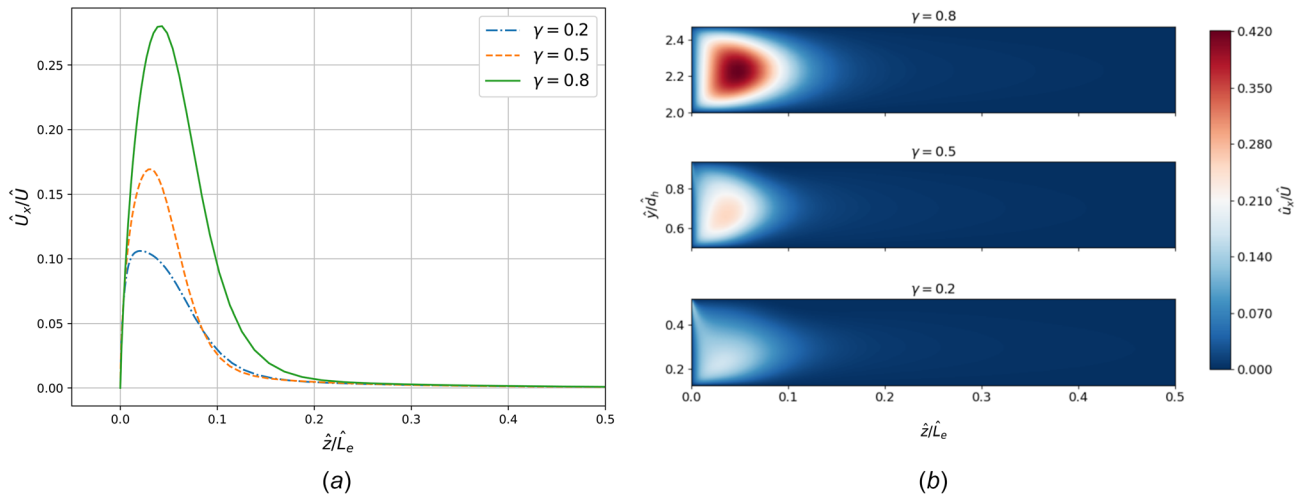


Fig. 7 Plots of azimuthal flow for Re = 200 and eccentricity ε=0.7. Plot (*a*) shows the normalized azimuthal bulk velocity, $\hat{U}_x/\hat{U}$, while plot (*b*) shows a contour plot of $\hat{u}_x/\hat{U}$ extracted from the same plane from which $\hat{U}_x/\hat{U}$ is calculated from. As a reminder to the reader, note that the difference between $\hat{U}_x$ and $\hat{u}_x$ is that $\hat{U}_x$ denotes the mean velocity summarized over the gap that is normal on the inner cylinder and perpendicular to the axis of eccentricity (see Fig. 2 for reference), while $\hat{u}_x$ is the velocity component extracted at any point on the same plane.

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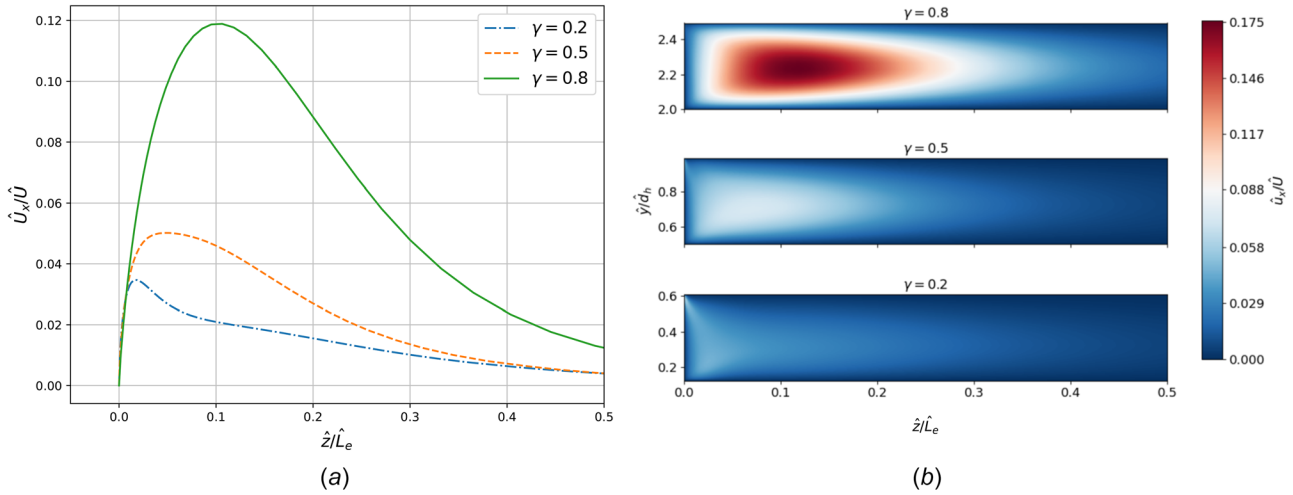


Fig. 8 Plots of azimuthal flow for Re = 200 and eccentricity ε=0.3. Plot (*a*) shows the normalized azimuthal bulk velocity, $\hat{U}_x/\hat{U}$, while plot (*b*) shows a contour plot of $\hat{u}_x/\hat{U}$.

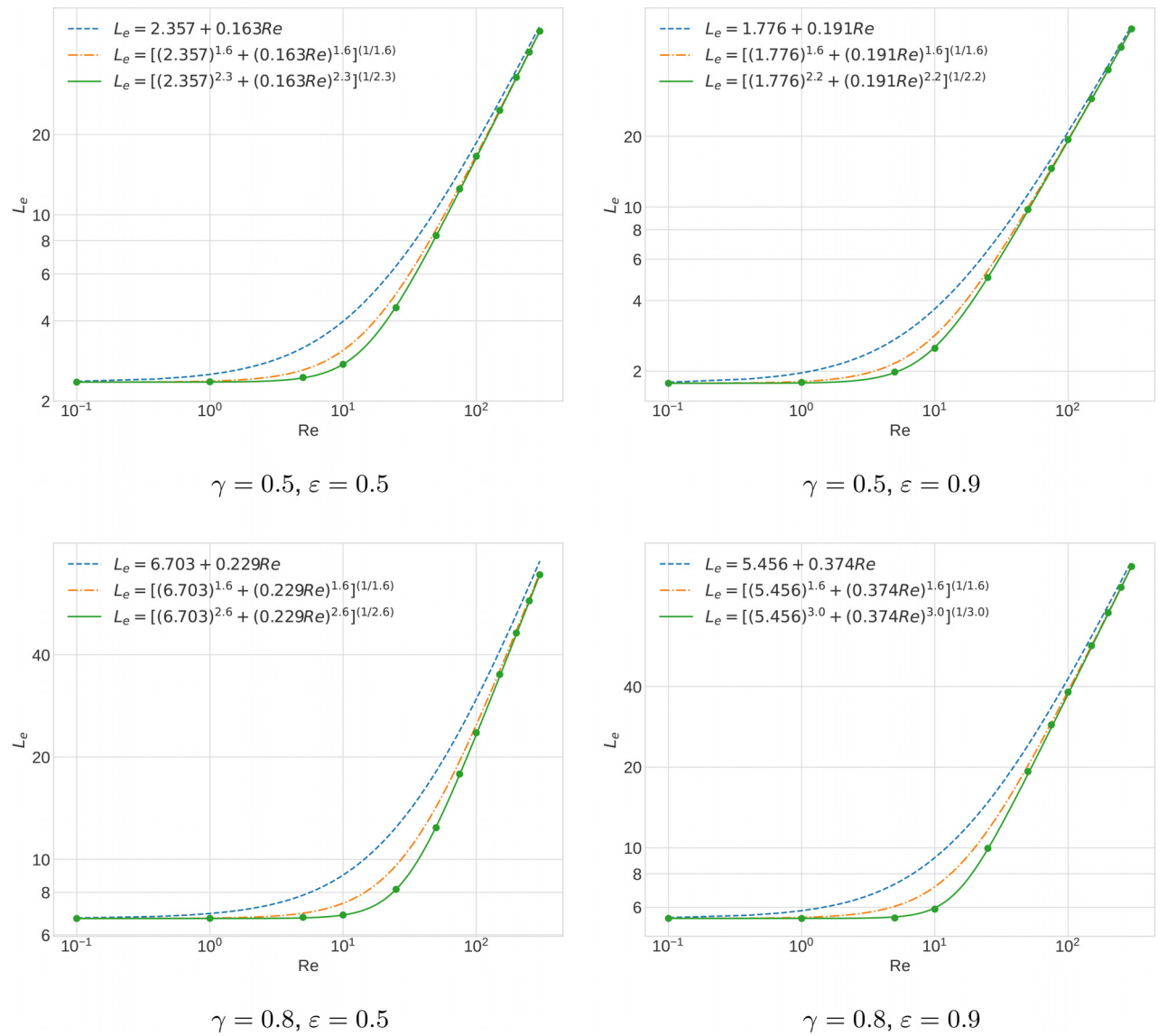


Fig. 9 Plot of entrance lengths and entrance length correlations for four different geometries. The plot includes the entrance lengths from each simulation (filled dots), and the correlation function defined in Eq. (3) with three different exponents *n*. These are *n* = 1.0 (dashed line), *n* = 1.6 (dash-dotted line), and the calculated exponent of the current study (solid line). (*a*) γ=0.5, ε=0.5; (*b*) γ=0.5, ε=0.9; (*c*) γ=0.8, ε=0.5; and (*d*) γ=0.8, ε=0.9.

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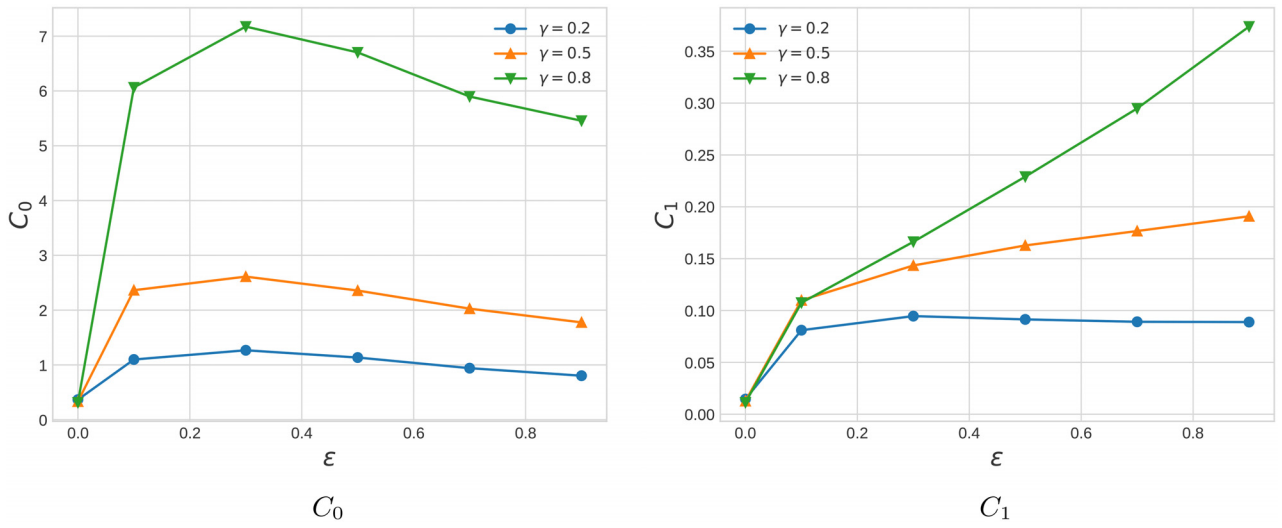


Fig. 10 Summary of the correlations C_0 and C_1 for entrance lengths as function of eccentricity for three different aspect ratios

was established for the dimensionless entrance length that scales with the mean dimensionless azimuthal distance between the wide and narrow sides of the annulus. For eccentric annuli in the inertial regime, an increased aspect ratio generally increases the entrance lengths. Additionally, larger aspect ratios were more sensitive to increased eccentricity, whereas annuli with the lowest aspect ratio were almost independent of the eccentricity. In terms of the nonlinear correlation function, using the same exponent as in previous studies on entrance lengths in pipe and channel flows ($n = 1.6$) was sufficient for concentric annuli in general and also suitable for the lowest aspect ratio ($\gamma = 0.2$) eccentric annuli. However, for the eccentric annuli with higher aspect ratios ($\gamma = 0.5$, $\gamma = 0.8$), significantly larger values of n were found. We attributed the increase of n to the larger peaks in azimuthal bulk flow taking place in the entrance region, caused by the redistribution of liquid from the narrow side to the wider side of the annuli. The improved correlations for entrance lengths in eccentric annuli can have significance in relation to conducting physical or numerical experiments when the fully developed flow is of interest.

6 Limitations and Future Work

A limitation of the current work is the lack of experimental validation. To the author's knowledge, no earlier experimental studies exist on entrance lengths in eccentric annuli for laminar flow. This gap in validation highlights the need for future experimental studies to verify the proposed entrance length correlations. Another limitation is the limited range of Reynolds numbers investigated. The focus on the laminar flow regime ensured no gap instabilities were present. We expect that gap instabilities and turbulence could shorten the entrance lengths significantly, and the proposed correlations presented here would likely be invalidated. A challenge with studying entrance lengths at higher Reynolds numbers is that no steady solution can be obtained, and so the definition of the entrance lengths must be altered to some mean quantity in that case. This constraint limits the direct applicability of our results to higher Reynolds number flows. Extending the current work to the turbulent regime would bridge a missing gap in the literature. Another limitation is the assumption of an idealized annular geometry. In an experimental study, obtaining an annulus with constant eccentricity in the axial direction could be challenging due to gravity. In the case of an annulus with nonconstant eccentricity, flow perturbations could arise in the flow and alter the obtained entrance lengths. Also, since the current study is based on a simple (constant viscosity) fluid model, realistic industrial fluids may exhibit significantly different flow characteristics and development lengths. This topic is however an interesting and relevant direction of future research.

Acknowledgment

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Funding Data

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

$\hat{c}$ = offset distance between axes of inner and outer pipes (m)
$\hat{C}$ = geometric constant in bipolar coordinate system (m)
C_0, C_1, n = correlation function coefficients (dimensionless)
$\hat{D}_H$ = annulus hydraulic diameter, $\hat{D}_o - \hat{D}_i$ (m)
$\hat{D}_i$ = annulus inner diameter (m)
$\hat{D}_o$ = annulus outer diameter (m)
E_r = error of the fully developed velocity relative to the semi-analytical solution (dimensionless)
g = growth rate parameter for computational mesh (dimensionless)
$\hat{L}_e$ = entrance length (m)
N_r, N_θ = number of cells in radial and azimuthal directions (dimensionless)
$\hat{p}$ = pressure (Pa)
$\langle \hat{p} \rangle$ = mean pressure (Pa)

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p_z = normalized pressure (dimensionless)
$R_{\max,c}$ = radial position of peak velocity (m)
$\hat{S}$ = mean azimuthal distance between wide and narrow side of annulus (m)
$\hat{\mathbf{u}}$ = fluid velocity vector (m/s)
$\hat{U}$ = imposed axial bulk velocity (m/s)
$\hat{u}_f$ = fully developed velocity profile (m/s)
$\hat{U}_x$ = transverse bulk velocity (m/s)
v = fully developed velocity profile (dimensionless)
$\Delta\hat{L}_{\max}, \Delta\hat{L}_{\min}$ = maximum and minimum axial length of computational cells (m)

Greek Symbols

α, β = coefficients for defining bipolar coordinate system
$\hat{\mu}$ = fluid viscosity (Pa·s)
ξ, η = bipolar coordinates
$\hat{\rho}$ = fluid mass density (kg/m^3)

Superscripts and Subscripts

b = bulk value
e = entrance length

Dimensionless Groups

L_e = nondimensional entrance length, $\hat{L}_e/\hat{D}_H$
Re = Reynolds number, $\hat{\rho}\hat{U}\hat{D}_H/\hat{\mu}$
S = mean azimuthal distance between wide and narrow side of annulus, $\hat{S}/\hat{D}_H$
γ = aspect ratio, $\hat{D}_i/\hat{D}_o$
ε = eccentricity, $2\hat{c}/\hat{D}_H$

Appendix: Fully Developed Laminar Velocity Profile for Eccentric Annulus

In the following, we provide the analytical result for the fully developed laminar velocity profile in an eccentric annulus, from the derivation of Snyder and Goldstein [13]. Their derivation was given in bipolar coordinates, which are related to rectangular coordinates by the following transformations [13,24]:

$$\hat{x} = \frac{\hat{C}\sinh\eta}{\cosh\eta - \cos\xi} \tag{A1a}$$

$$\hat{y} = \frac{\hat{C}\sin\xi}{\cosh\eta - \cos\xi} \tag{A1b}$$

Here, the geometrical constant $\hat{C}$ is defined as $\hat{C} = \hat{R}_i \sinh\alpha = \hat{R}_o \sinh\beta$, with α and β denoting the lines of constant η that coincides with the inner and outer walls of the eccentric annulus, as shown in Fig. 11. These parameters are given by

$$\cosh\alpha = \frac{1}{\gamma}\frac{\gamma(1+\varepsilon^2) + (1-\varepsilon^2)}{2\varepsilon} \tag{A2a}$$

$$\cosh\beta = \frac{\gamma(1-\varepsilon^2) + (1+\varepsilon^2)}{2\varepsilon} \tag{A2b}$$

The velocity profile derived in Ref. [13] can be expressed as

$$\hat{u}_f(\eta,\xi) = \frac{\hat{C}^2}{\hat{\mu}}\left(-\frac{\mathrm{d}\hat{p}}{\mathrm{d}\hat{z}}\right) v(\eta,\xi) \tag{A3}$$

where $v(\eta,\xi)$ is a purely geometric factor

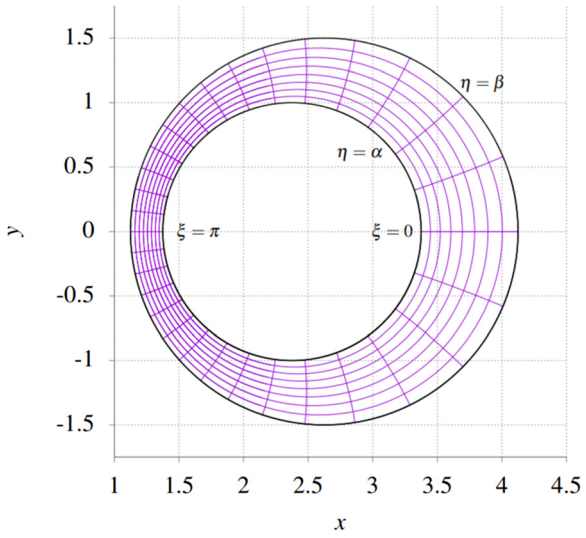


Fig. 11 In bipolar coordinates (η, ξ), the inner and outer walls of the eccentric annulus correspond to curves with fixed $\eta=\alpha$ and $\eta=\beta$, respectively, and $\xi\in[0, 2\pi]$. The relation between rectangular and bipolar coordinates is here illustrated for $\gamma=2/3$ and $\varepsilon=1/2$, and an inner pipe of unity radius.

$$v(\eta,\xi) = F + E\eta - \frac{1}{2}\operatorname{ctnh}\eta + \sum_{n=1}^{\infty}\left\{A_n \mathrm{e}^{n\eta} + (B_n - \operatorname{ctnh}\eta)\mathrm{e}^{-n\eta}\right\}\cos n\xi \tag{A4a}$$

and where

$$F = \frac{\alpha\operatorname{ctnh}\beta - \beta\operatorname{ctnh}\alpha}{2(\alpha-\beta)} \tag{A4b}$$

$$E = \frac{\operatorname{ctnh}\alpha - \operatorname{ctnh}\beta}{2(\alpha-\beta)} \tag{A4c}$$

$$A_n = \frac{\operatorname{ctnh}\alpha - \operatorname{ctnh}\beta}{\mathrm{e}^{2n\alpha} - \mathrm{e}^{2n\beta}} \tag{A4d}$$

$$B_n = \frac{\mathrm{e}^{2n\alpha}\operatorname{ctnh}\beta - \mathrm{e}^{2n\beta}\operatorname{ctnh}\alpha}{\mathrm{e}^{2n\alpha} - \mathrm{e}^{2n\beta}} \tag{A4e}$$

References

[1] Kelessidis, V. C., Maglione, R., and Mitchell, R. F., 2011, "Drilling Hydraulics," *Fundamentals of Drilling Engineering*, R. F. Mitchell, and S. Z. Miska, eds., Society of Petroleum Engineers, Richardson, TX, pp. 179–310.
[2] Batra, R., and Jena, B., 1990, "Entrance Region Flow of Blood in Concentric Annulus," Int. J. Eng. Sci., **28**(5), pp. 407–419.
[3] Choueiri, G. H., and Tavoularis, S., 2014, "Experimental Investigation of Flow Development and Gap Vortex Street in an Eccentric Annular Channel. Part 1. Overview of the Flow Structure," J. Fluid Mech., **752**, pp. 521–542.
[4] Velusamy, K., and Garg, V. K., 1994, "Entrance Flow in Eccentric Annular Ducts," Int. J. Numer. Methods Fluids, **19**(6), pp. 493–512.
[5] White, F. M., 1991, *Viscous Fluid Flow*, 2nd ed., McGraw-Hill, New York.
[6] Osiptsov, A. A., and Asmolov, E. S., 2008, "Asymptotic Model of the Inertial Migration of Particles in a Dilute Suspension Flow Through the Entry Region of a Channel," Phys. Fluids, **20**(12), p. 123301.
[7] Joshi, Y., and Vinoth, B. R., 2018, "Entry Lengths of Laminar Pipe and Channel Flows," ASME J. Fluids Eng., **140**(6), p. 061203.
[8] Durst, F., Ray, S., Ünsal, B., and Bayoumi, O. A., 2005, "The Development Lengths of Laminar Pipe and Channel Flows," ASME J. Fluids Eng., **127**(6), pp. 1154–1160.
[9] Poole, R. J., 2010, "Development-Length Requirements for Fully Developed Laminar Flow in Concentric Annuli," ASME J. Fluids Eng., **132**(6), p. 064501.
[10] Poole, R. J., and Ridley, B. S., 2007, "Development-Length Requirements for Fully Developed Laminar Pipe Flow of Inelastic Non-Newtonian Liquids," ASME J. Fluids Eng., **129**(10), pp. 1281–1287.
[11] Poole, R. J., and Chhabra, R. P., 2010, "Development Length Requirements for Fully Developed Laminar Pipe Flow of Yield Stress Fluids," ASME J. Fluids Eng., **132**(3), p. 034501.

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[12] Heyda, J. F., 1959, "A Green's Function Solution for the Case of Laminar Incompressible Flow Between Non-Concentric Circular Cylinders," J. Franklin Inst., **267**(1), pp. 25–34.
[13] Snyder, W. T., and Goldstein, G. A., 1965, "An Analysis of Fully Developed Laminar Flow in an Eccentric Annulus," AIChE J., **11**(3), pp. 462–467.
[14] Feldman, E. E., Hornbeck, R. W., and Osterle, J. F., 1982, "A Numerical Solution of Laminar Developing Flow in Eccentric Annular Ducts," Int. J. Heat Mass Transfer, **25**(2), pp. 231–241.
[15] Lamarche-Gagnon, M.-E., and Tavoularis, S., 2021, "Further Experiments and Analysis on Flow Instability in Eccentric Annular Channels," J. Fluid Mech., **915**, p. A34.
[16] Mewis, J., and Wagner, N. J., 2012, *Colloidal Suspension Rheology*, Cambridge University Press, Cambridge, UK.
[17] Walton, I. C., and Bittleston, S. H., 1991, "The Axial Flow of a Bingham Plastic in a Narrow Eccentric Annulus," J. Fluid Mech., **222**, pp. 39–60.
[18] Larson, R. G., and Wei, Y., 2019, "A Review of Thixotropy and Its Rheological Modeling," J. Rheol., **63**(3), pp. 477–501.
[19] Corvisier, P., Nouar, C., Devienne, R., and Lebouché, M., 2001, "Development of a Thixotropic Fluid Flow in a Pipe," Exp. Fluids, **31**(5), pp. 579–587.
[20] Barnes, H. A., 1995, "A Review of the Slip (Wall Depletion) of Polymer Solutions, Emulsions and Particle Suspensions in Viscometers: Its Cause, Character, and Cure," J. Non-Newtonian Fluid Mech., **56**(3), pp. 221–251.
[21] Shah, R., and London, A., 1978, "Chapter III—Dimensionless Groups and Generalized Solutions," *Laminar Flow Forced Convection in Ducts*, R. Shah, and A. London, eds., Academic Press, Cambridge, MA, pp. 37–60.
[22] Churchill, S. W., and Usagi, R., 1972, "A General Expression for the Correlation of Rates of Transfer and Other Phenomena," AIChE J., **18**(6), pp. 1121–1128.
[23] Barker, J. W., Powers, J., and Mitchell, R. F., 2011, "Casing Design," *Fundamentals of Drilling Engineering*, R. F. Mitchell, and S. Z. Miska, eds., Society of Petroleum Engineers, Richardson, TX, pp. 385–448.
[24] Bird, R. B., Armstrong, R. C., and Hassager, O., 1987, *Dynamics of Polymeric Liquids*, 2nd ed., Vol. 1, Wiley, Hoboken, NJ.

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